
Lecture 6: Static Games
Author: Daniel Jaramillo Calderon
2026

Contents

1 Strategic Decision Making	2
2 Strategic Form Games	2
3 Dominant Strategies	3
4 Nash Equilibrium	4
4.1 Best Responses	4
4.2 Examples	4
5 Mixed Strategies	5
5.1 Two Properties of Nash Equilibria	6
5.2 Finding Mixed Equilibria: Best-Response Correspondence	6
6 Cournot Nash Equilibrium: Worked Solution	7
7 Applications	8
7.1 Bertrand Competition with Identical Goods	8
7.2 Hotelling's Spatial Differentiation	9
8 Incomplete Information and Bayesian Games	9

1. Strategic Decision Making

Game theory studies how rational agents behave in situations where each must first analyze the choices of the others in order to know which decision is best for himself. We will study two game forms:

- **Strategic-form (static) games:** players make a single, simultaneous choice.
- **Extensive-form (dynamic) games:** players choose in sequence (Lecture 7).

Non-strategic decisions can be analyzed in isolation: given prices and income, a consumer chooses a basket. **Strategic** decisions are different — the optimal action depends on what others do. The penalty kick is a canonical example:

	<i>L</i>	<i>R</i>
<i>U</i>	3, 0	0, -4
<i>D</i>	2, 4	-1, 8

Each player seeks to maximize his payoff by *strategically reasoning*: he knows what is best for the others and decides accordingly. In some games optimal play requires being *unpredictable*.

2. Strategic Form Games

We focus first on games with **complete information** (payoffs observable; players know which world they live in) and **perfect information** (actions observable), in which players choose strategies simultaneously.

A **strategic-form game** is a tuple

$$G = (S_i, u_i)_{i=1}^N,$$

where for each player i :

- S_i is the (finite or infinite) set of pure strategies available to player i .
- $u_i : S \rightarrow \mathbb{R}$ maps a joint strategy profile $s = (s_1, \dots, s_N) \in S \equiv \prod_j S_j$ into player i 's payoff.

We write s_{-i} for an element of $S_{-i} \equiv \prod_{j \neq i} S_j$ — the strategies of everyone except i .

Example

Penalty kick. $i \in \{1, 2\}$, $S_i = \{L, R\}$, payoffs as above.

Example

Cournot duopoly. Two firms choose quantities simultaneously:

$$i \in \{1, 2\}, \quad S_i = \mathbb{R}_+, \quad u_i(q_1, q_2) = P(q_1 + q_2) q_i - c q_i.$$

The goal of solving a game is not to predict that all agents play the solution but to identify outcomes that are *stable under rationality*: scenarios where no player has an incentive to deviate.

3. Dominant Strategies

A first attempt at narrowing down predictions uses the idea of *preferable* strategies. No beliefs about opponents are needed.

$\hat{s}_i \in S_i$ is **strictly dominant** for player i if

$$u_i(\hat{s}_i, s_{-i}) > u_i(s_i, s_{-i}) \quad \forall (s_i, s_{-i}) \in S, \quad s_i \neq \hat{s}_i.$$

Strictly dominant strategies are rare: most games have none. Consider

	L	M	R
U	3, 0	0, -5	0, -4
C	1, -1	3, 3	-2, 4
D	2, 4	4, 1	-1, 8

no player has a strictly dominant strategy. But some strategies are *never* best: for player 1, $u_1(C, \cdot) < u_1(D, \cdot)$ uniformly; for player 2, $u_2(\cdot, R) > u_2(\cdot, M)$ uniformly.

$s_i \in S_i$ is **strictly dominated** by $s'_i \in S_i$ if

$$u_i(s'_i, s_{-i}) > u_i(s_i, s_{-i}) \quad \forall s_{-i} \in S_{-i}.$$

IESDS. A rational player never plays a strictly dominated strategy. *Iterated elimination of strictly dominated strategies* (IESDS) repeatedly removes such strategies. The procedure relies on

- rationality (R): no player plays a strictly dominated action;
- knowledge of rationality ($K(R)$): players know others won't either;
- $K^2(R), K^3(R), \dots$: knowledge of knowledge of rationality.

In the matrix above, eliminating C and M yields a smaller game with a strictly dominant strategy.

$\hat{s}_i$ **weakly dominates** s_i if $u_i(\hat{s}_i, s_{-i}) \geq u_i(s_i, s_{-i})$ for all s_{-i} , with at least one strict inequality.

Example

D weakly dominated for player 1 and R weakly dominated for player 2 in:

	L	R
U	1, 1	0, 0
D	0, 0	0, 0

The game is dominance-solvable: (U, L) is the unique rationality-consistent outcome.

4. Nash Equilibrium

Most games are not dominance solvable, and many do not even have a strictly dominated strategy. We need a stronger solution concept.

A joint strategy $s^* \in S$ is a **Nash equilibrium** of $G = (S_i, u_i)_{i=1}^N$ if for every player i and every $s_i \in S_i$,

$$u_i(s^*) \geq u_i(s_i, s_{-i}^*).$$

Informally, no player can do strictly better by unilaterally changing his strategy.

4.1. Best Responses

$$BR_i(s_{-i}) \equiv \arg \max_{s_i \in S_i} u_i(s_i, s_{-i}).$$

A Nash equilibrium is a profile in which each player's strategy is a best response to the others.

4.2. Examples

Example

Prisoner's Dilemma.

	N	C
N	-1, -1	-3, 0
C	0, -3	-2, -2

$BR_i(N) = C$ and $BR_i(C) = C$: (C, C) is the unique Nash equilibrium, even though (N, N) Pareto-dominates it.

Example

Coordination. The game below has *two* pure-strategy Nash equilibria: (U, L) and (D, R) .

	L	R
U	1, 1	0, 0
D	0, 0	1, 1

Example

Matching Pennies / Penalty Kick.

	L	R
L	-1, 1	1, -1
R	1, -1	-1, 1

No pure-strategy Nash equilibrium exists. We need to enlarge the strategy space.

5. Mixed Strategies

Suppose both the shooter and goalkeeper privately flip a fair coin and play L if heads, R if tails. Given the method by which the opponent makes his choice, neither can do better by changing his own method. This is the idea behind mixed strategies.

A **mixed strategy** m_i for player i is a probability distribution over S_i :

$$m_i : S_i \rightarrow [0, 1], \quad \sum_{s_i \in S_i} m_i(s_i) = 1.$$

We denote the set of mixed strategies for player i by M_i . The pure strategies are the degenerate elements of M_i ($S_i \subset M_i$).

Since players randomize independently, the expected payoff under a profile $m = (m_1, \dots, m_N)$ is

$$u_i(m) = \sum_{s \in S} m_1(s_1) m_2(s_2) \cdots m_N(s_N) u_i(s).$$

$m^* \in M$ is a **Nash equilibrium** if for every i and every $m_i \in M_i$:

$$u_i(m^*) \geq u_i(m_i, m_{-i}^*).$$

Interpretation (population frequencies). Probabilities can be read as population shares choosing each action. The frequency evolves by imitation and reproduction; Nash equilibria correspond to

stationary distributions.

5.1. Two Properties of Nash Equilibria

If s_i and s'_i are both played with positive probability under m^* , then

$$u_i(s_i, m_{-i}^*) = u_i(s'_i, m_{-i}^*) = u_i(m^*).$$

Proof. Suppose s_i, s'_i are the only strategies played with positive probability. Then

$$u(m^*) = m^*(s_i) u(s_i, m_{-i}^*) + (1 - m^*(s_i)) u(s'_i, m_{-i}^*).$$

If $u(s_i, m_{-i}^*) > u(s'_i, m_{-i}^*)$, player i would strictly prefer the pure strategy $m^*(s_i) = 1$. ■

$$u(m_i^*, m_{-i}^*) \geq u(s_i, m_{-i}^*) \text{ for every } s_i \in S_i.$$

Every finite game has at least one Nash equilibrium in mixed strategies.

5.2. Finding Mixed Equilibria: Best-Response Correspondence

A systematic way to find mixed equilibria is to plot each player's *best-response correspondence* and look for intersections. Let $q = m_2(L)$ be the probability player 2 places on L .

Example

Matching Pennies.

	$L (q)$	$R (1 - q)$
L	$-1, 1$	$1, -1$
R	$1, -1$	$-1, 1$

Player 1's expected payoffs as a function of q :

$$u_1(L) = (-1)q + (1)(1 - q) = 1 - 2q,$$

$$u_1(R) = (1)q + (-1)(1 - q) = 2q - 1.$$

Player 1 prefers L if $q < \frac{1}{2}$, prefers R if $q > \frac{1}{2}$, and is indifferent at $q = \frac{1}{2}$.

Best-response correspondence $BR_1(q)$:

$$BR_1(q) = \begin{cases} \{L\} & q < \frac{1}{2}, \\ [0, 1] & q = \frac{1}{2}, \\ \{R\} & q > \frac{1}{2}. \end{cases}$$

By symmetry, $BR_2(p) = \{R\}$ if $p < \frac{1}{2}$, $[0, 1]$ if $p = \frac{1}{2}$, $\{L\}$ if $p > \frac{1}{2}$ (where $p = m_1(L)$). The two correspondences intersect only at $m^* = (p^*, q^*) = (\frac{1}{2}, \frac{1}{2})$. Both players earn 0 in expectation.

Example

Battle of the Sexes.

	$LX (q)$	$MW (1 - q)$
LX	2, 1	0, 0
MW	0, 0	1, 2

Player 1's indifference ($p = m_1(LX)$): $u_1(LX) = 2q = u_1(MW) = 1 - q \Rightarrow q^* = \frac{1}{3}$.

Player 2's indifference: $u_2(LX) = p = u_2(MW) = 2(1 - p) \Rightarrow p^* = \frac{2}{3}$.

The game has **three** Nash equilibria: two pure, (LX, LX) and (MW, MW) , and one mixed, $m^* = (p^* = \frac{2}{3}, q^* = \frac{1}{3})$.

Example

Reporting a Crime. n individuals witness a crime. Reporting costs $c > 0$; everyone benefits $v > c$ if at least one reports. Let p_i denote the probability that i reports.

Pure-strategy equilibria. Three exhaustive cases:

- Nobody reports $(N, N, \dots, N)$: any single deviator earns $v - c > 0$. Not an equilibrium.
- Two or more report: a reporter who switches to N still gets v , so deviation is profitable. Not an equilibrium.
- Exactly one reports, say agent i : i 's deviation gives $0 < v - c$; any other agent's deviation from N gives $v - c < v$. *Equilibrium.* There are n such pure equilibria, one for each potential reporter.

Mixed equilibrium. Indifference between R and N :

$$v - c = v(1 - \Pr(\text{no one else reports})) \implies \frac{c}{v} = \prod_{j \neq i} (1 - p_j).$$

6. Cournot Nash Equilibrium: Worked Solution

The Cournot duopoly was introduced in Section 2 as a canonical strategic-form game with continuous strategies. Here we solve it in full.

Example

Setup. Two firms simultaneously choose quantities $q_1, q_2 \geq 0$. Inverse demand is $P(Q) = a - Q$ with $Q = q_1 + q_2$ and $c = 0$ (no costs). Each firm i maximises

$$u_i(q_i, q_j) = (a - q_i - q_j) q_i.$$

Best-response function. FOC with respect to q_i :

$$a - 2q_i - q_j = 0 \implies \text{BR}_i(q_j) = \frac{a - q_j}{2}.$$

Nash equilibrium. By symmetry $q_1^* = q_2^* = q^*$:

$$q^* = \frac{a - q^*}{2} \implies q^* = \frac{a}{3}.$$

Total output $Q^* = 2a/3$ and price $P^* = a/3$.

Social optimum. Maximising $(a - Q)Q$ gives $Q^{**} = a/2 > Q^*$. Cournot output is below the efficient level — each firm ignores the negative externality it imposes on the other.

7. Applications

7.1. Bertrand Competition with Identical Goods

Two firms simultaneously set prices $p_1, p_2 \in \mathbb{R}_+$. Goods are identical and consumers buy from the cheaper firm (ties split the market). Demand $Q(p) = \max\{A - p, 0\}$ and marginal cost $c \in (0, A)$ are common.

$$u_i(p_1, p_2) = \begin{cases} Q(p_i)(p_i - c), & p_i < p_j, \\ \frac{1}{2}Q(p_i)(p_i - c), & p_i = p_j, \\ 0, & p_i > p_j. \end{cases}$$

Monopoly benchmark. $\pi^M(p) = (A - p)(p - c)$ peaks at $p^M = (A + c)/2$.

Best responses. Given rival's p_j :

1. $p_j > p^M$: firm sets p^M .
2. $c < p_j \leq p^M$: undercut by an infinitesimal amount. No exact best response in the continuum.
3. $p_j = c$: any $p \geq c$ yields zero profit.
4. $p_j < c$: stay out.

The unique Nash equilibrium is $p_1 = p_2 = c$. Both firms earn zero profit despite duopoly. Price equals marginal cost as in perfect competition.

Underlying assumptions: identical products; no capacity constraints; full consumer information.

7.2. Hotelling's Spatial Differentiation

Firms sell the same good but are spatially differentiated.

- Firm 1 located at $x = 0$, firm 2 at $x = 1$.
- Consumers uniformly along $[0, 1]$, each buying one unit, with travel cost $t > 0$ per unit distance.
- Consumer at x values the good at $v > c$; her utilities are $v - p_1 - tx$ and $v - p_2 - t(1 - x)$.
- The consumer-location distribution has CDF F symmetric around $\frac{1}{2}$ with density f .

Indifferent consumer.

$$x^I = \frac{1}{2} + \frac{p_2 - p_1}{2t}.$$

Market shares and profits.

$$S_1(p) = F(x^I), \quad S_2(p) = 1 - F(x^I), \quad u_i(p) = (p_i - c) S_i(p).$$

Symmetric equilibrium. Imposing $p_1 = p_2 = p^*$:

$$p^* = c + \frac{t}{f(\frac{1}{2})}.$$

Comparative statics.

- Higher t : more differentiation, higher markup.
- Higher $f(\frac{1}{2})$: more “disloyal” consumers near the midpoint, lower markup.
- $t \rightarrow 0$ or $f(\frac{1}{2}) \rightarrow \infty$: $p^* \rightarrow c$ (Bertrand outcome).

Uniform special case. With $F(x) = x$ we get $p^* = c + t$. The profit function is concave in p_i , confirming this is a Nash equilibrium.

8. Incomplete Information and Bayesian Games

Players may not know each other's payoffs (e.g. a firm's marginal cost). We formalize this by introducing *types*.

Definition 8.1 — Bayesian Game

A **game of incomplete information** is a tuple $G = (p_i, T_i, S_i, u_i)_{i=1}^N$, where

- T_i is the (finite) set of possible types for player i ;
- $u_i : S \times T \rightarrow \mathbb{R}$ specifies i 's payoff for every strategy profile and type profile;

- $p_i(\cdot | t_i)$ is a probability distribution on T_{-i} representing i 's beliefs about others' types given his own type t_i .

Strategies vs. actions. A pure strategy is a contingent plan: $s_i : T_i \rightarrow S_i$. The realized *action* depends on the type observed.

Strategic-form representation. For each i and each $t_i \in T_i$, treat the pair (i, t_i) as a distinct player. The associated strategic-form game has payoffs

$$v_{t_i}(s) = \sum_{t_{-i} \in T_{-i}} p_i(t_{-i} | t_i) u_i(s_1(t_1), \dots, s_N(t_N); t_1, \dots, t_N).$$

A Bayesian–Nash equilibrium of G is a Nash equilibrium of its associated strategic-form game.

Example

Cournot with incomplete information. Firm 1's marginal cost is c_1 , common knowledge. Firm 2 is of two types: $c_2 \in \{\underline{c}, \bar{c}\}$ with $\Pr(c_2 = \bar{c}) = p$. Firm 1 chooses one quantity q_1 . Firm 2 chooses two contingent quantities $(\underline{q}, \bar{q})$ depending on its type. The Bayesian–Nash equilibrium is found by treating $\bar{2}$ and $\underline{2}$ as separate players whose payoffs are appropriately weighted by p and $1 - p$.

Example

Bertrand with private cost. Firm 1 has cost 0 (known). Firm 2 has cost $c \in \{1, 4\}$, each with probability $\frac{1}{2}$. Prices restricted to $\{1, 4, 6\}$; market demand is $8 - p$. The strategic form replaces firm 2 with two type-players (low cost and high cost), each of whom independently picks a price; firm 1's payoff is the expectation over firm 2's type. A Bayesian–Nash equilibrium is a Nash equilibrium of this enlarged matrix game.

Reference: Jehle & Reny, Advanced Microeconomic Theory, Chapter 7.